

Air University



Operating Systems (Lab)

Year 2026

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Academic Year: 2024 – 2028

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BS – Cyber Security – Fall – 24 (A)

Date of Submission: February 20, 2026

Lab Task # 03

Task Scenario:

Create a directory on the Desktop:

```
usama@ubuntu:~$ ls
Desktop Documents Downloads Music Pictures Public snap Templates Videos
usama@ubuntu:~$ cd Desktop
usama@ubuntu:~/Desktop$ mkdir secure_project
```

In the "secure_project" directory, three files and two folders were created.

```
usama@ubuntu:~/Desktop$ cd secure_project
usama@ubuntu:~/Desktop/secure_project$ touch report.txt data.txt run.sh
usama@ubuntu:~/Desktop/secure_project$ mkdir private_folder shared_folder
usama@ubuntu:~/Desktop/secure_project$ ls -l
total 8
-rw-rw-r-- 1 usama usama 0 Feb 20 10:01 data.txt
drwxrwxr-x 2 usama usama 4096 Feb 20 10:01 private_folder
-rw-rw-r-- 1 usama usama 0 Feb 20 10:01 report.txt
-rw-rw-r-- 1 usama usama 0 Feb 20 10:01 run.sh
drwxrwxr-x 2 usama usama 4096 Feb 20 10:01 shared_folder
```

Task Requirements:

1. Script protection

```
usama@ubuntu:~/Desktop/secure_project$ chmod 764 run.sh
usama@ubuntu:~/Desktop/secure_project$ ls -l run.sh
-rwxrw-r-- 1 usama usama 0 Feb 20 10:01 run.sh
usama@ubuntu:~/Desktop/secure_project$ ./run.sh
usama@ubuntu:~/Desktop/secure_project$
```

2. Confidential report

```
usama@ubuntu:~/Desktop/secure_project$ chmod g-w,o-r report.txt
usama@ubuntu:~/Desktop/secure_project$ ls -l report.txt
-rw-r----- 1 usama usama 0 Feb 20 10:01 report.txt
```

3. Highly secure directory

```
usama@ubuntu:~/Desktop/secure_project$ chmod 700 private_folder
usama@ubuntu:~/Desktop/secure_project$ ls -l private_folder
total 0
usama@ubuntu:~/Desktop/secure_project$ ls -l
total 8
-rw-rw-r-- 1 usama usama 0 Feb 20 10:01 data.txt
drwx----- 2 usama usama 4096 Feb 20 10:01 private_folder
-rw-r----- 1 usama usama 0 Feb 20 10:01 report.txt
-rwxrw-rw- 1 usama usama 0 Feb 20 10:01 run.sh
drwxrwxr-x 2 usama usama 4096 Feb 20 10:01 shared_folder
```

➤ **Can someone list the files inside it?**

No one can list the files in private_folder; only the owner has permission.

➤ **Can someone enter it?**

No one can enter the private_folder because all permissions have been revoked.

4. Shared workspace:

```
usama@ubuntu:~/Desktop/secure_project$ chmod 775 shared_folder
usama@ubuntu:~/Desktop/secure_project$ ls -l
total 8
-rw-rw-r-- 1 usama usama 0 Feb 20 10:01 data.txt
drwx----- 2 usama usama 4096 Feb 20 10:01 private_folder
-rw-r----- 1 usama usama 0 Feb 20 10:01 report.txt
-rwxrw-r-- 1 usama usama 0 Feb 20 10:01 run.sh
drwxrwxr-x 2 usama usama 4096 Feb 20 10:01 shared_folder
```

5. Dangerous configuration check:

```
usama@ubuntu:~/Desktop/secure_project$ chmod 777 data.txt
usama@ubuntu:~/Desktop/secure_project$ ls -l data.txt
-rwxrwxrwx 1 usama usama 0 Feb 20 10:01 data.txt
```

➤ **Explain why this is risky?**

Granting all permissions to everyone is risky because anyone can manipulate the file.

➤ **Who can modify the file now?**

Anyone from the group or outside the group can modify the file because all permissions are granted.

Restore it to secure mode:

```
usama@ubuntu:~/Desktop/secure_project$ chmod 664 data.txt
usama@ubuntu:~/Desktop/secure_project$ ls -l data.txt
-rw-rw-r-- 1 usama usama 0 Feb 20 10:01 data.txt
```

6. Ownership test

```
usama@ubuntu:~/Desktop/secure_project$ sudo chown root data.txt
[sudo] password for usama:
usama@ubuntu:~/Desktop/secure_project$ ls -l
total 8
-rw-rw-r-- 1 root usama 0 Feb 20 10:01 data.txt
drwx----- 2 usama usama 4096 Feb 20 10:01 private_folder
-rw-r----- 1 usama usama 0 Feb 20 10:01 report.txt
-rwxrw-r-- 1 usama usama 0 Feb 20 10:01 run.sh
drwxrwxr-x 2 usama usama 4096 Feb 20 10:01 shared_folder
```

➤ **Can the original user edit it?**

No, as long as the user is a standard non-root user and the file permissions are set to default.

➤ **Why?**

Changing the file owner to root demotes the original user to "Other," which lacks write permissions under the default security settings (644), effectively locking the file.

THE END
